

# Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

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## Supplementary Information

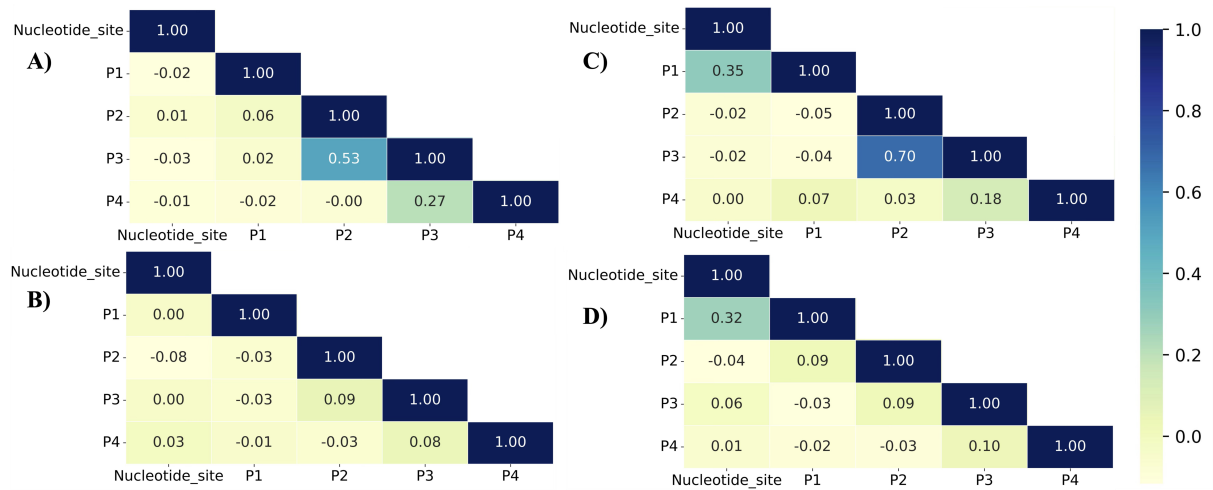


Figure S1: Pearson Correlation for the volume fluctuations among different pockets and the effect of nucleotide binding. **A)** GDP-bound Rac1; **B)** GTP-bound Rac1; **C)** Nucleotide-free state created by deleting GDP from Rac1; **D)** Nucleotide-free state created by deleting GTP from Rac1.

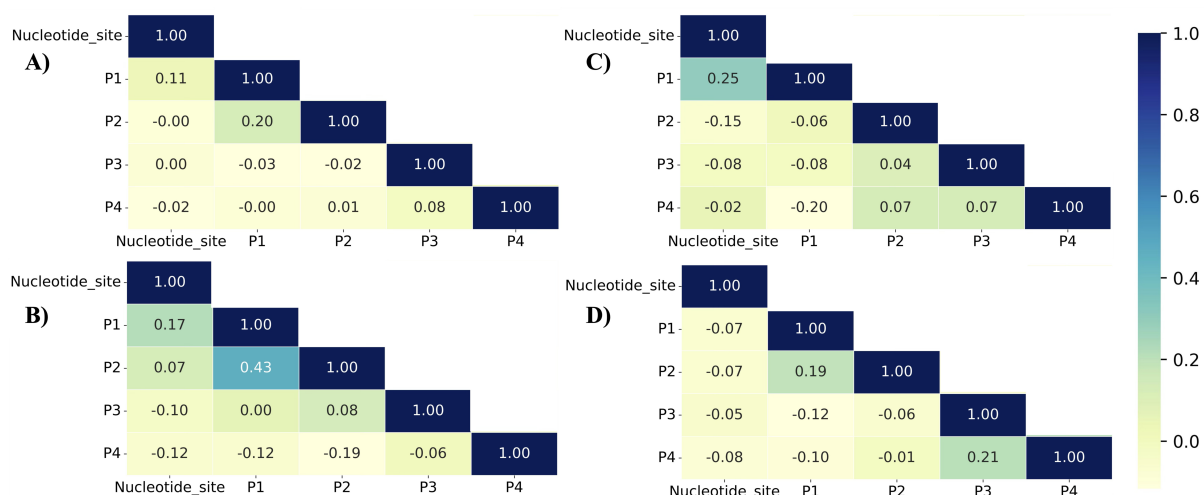


Figure S2: Pearson Correlation for the volume fluctuations among different pockets and the effect of GEF binding. **A)** PREX1-bound Rac1; **B)** TIAM1-bound Rac1; **C)** VAV1-bound Rac1; **D)** DOCK2-bound Rac1. Rac1 in all GEF-bound systems is in the nucleotide-free state.

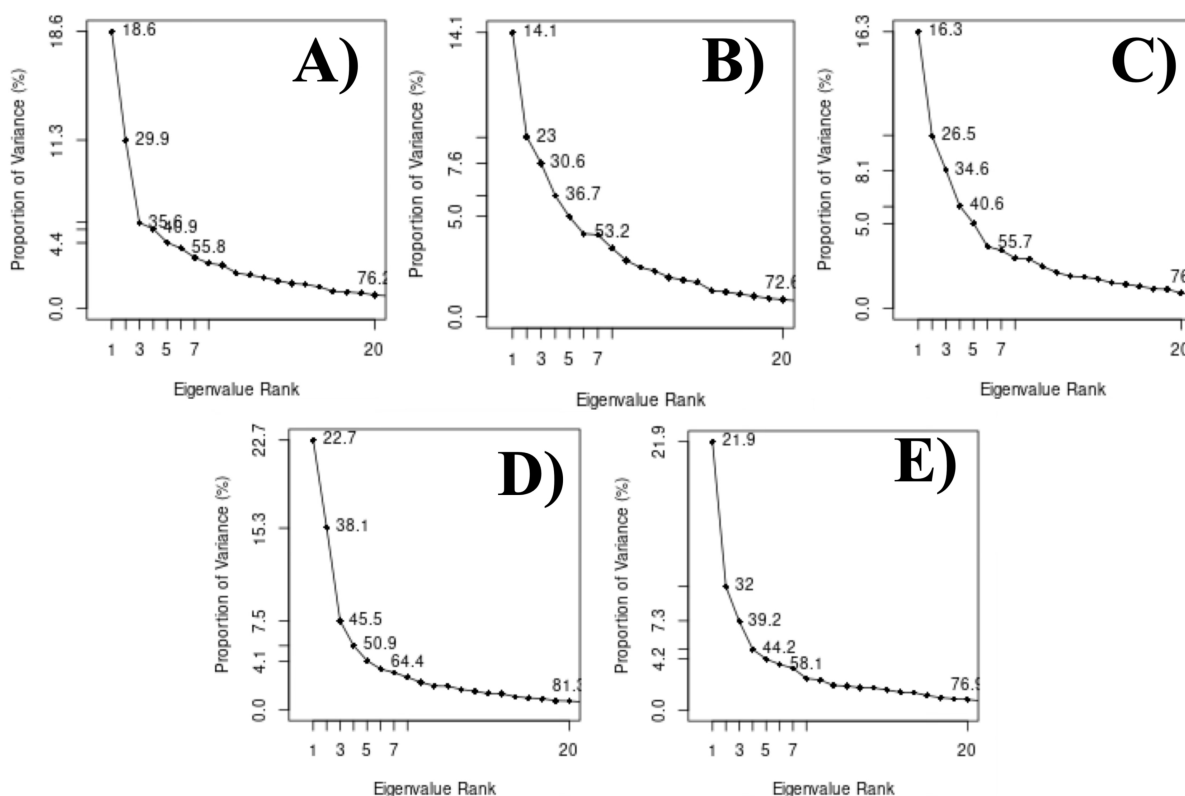


Figure S3: Scree plot to compute the cumulative variance explained by principal components. **A)** *apo*-Rac1 in TIP3P water box; **B)** GTP-bound Rac1 in TIP3P water box; **C)** *apo*-Rac1 in TIP3P water+isopropanol box; **D)** *apo*-Rac1 in TIP3P water+phenol box; **E)** *apo*-Rac1 in TIP3P water+toluene box.

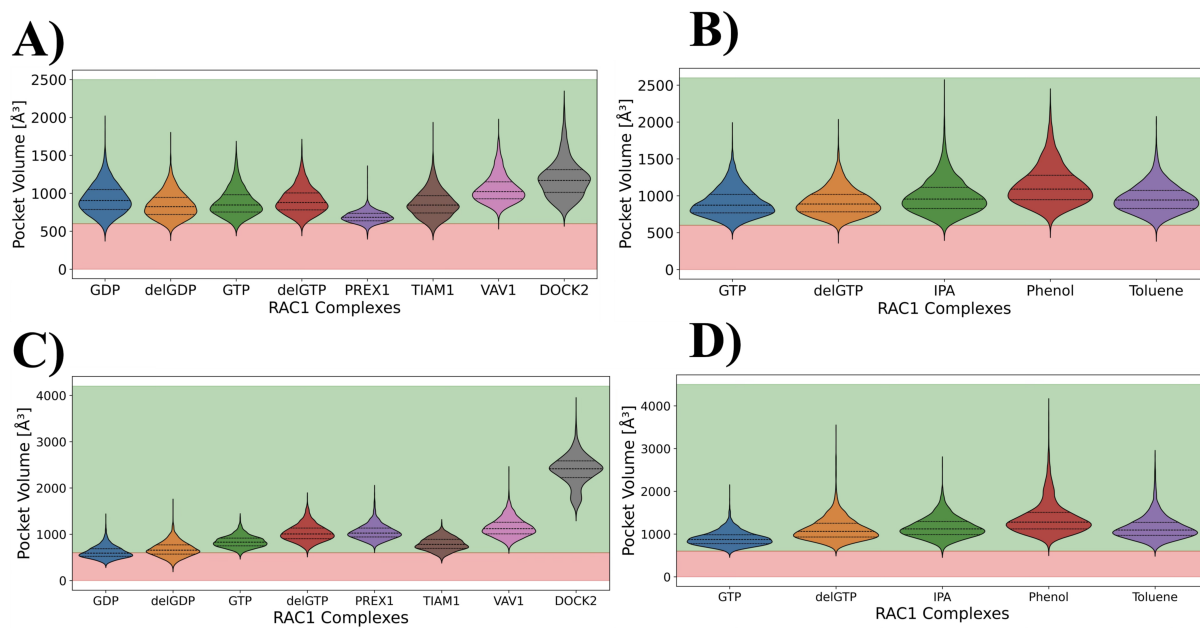


Figure S4: Analysis of pocket volume changes by combining pockets P1 with P2, and P3 with P4. **A)** Combined pocket volume analysis for site P1+P2 ; **B)** Combined pocket volume analysis for site P1+P2 in co-solvent; **C)** Combined pocket volume analysis for site P3+P4; **D)** Combined pocket volume analysis for site P3+P4 in co-solvent.

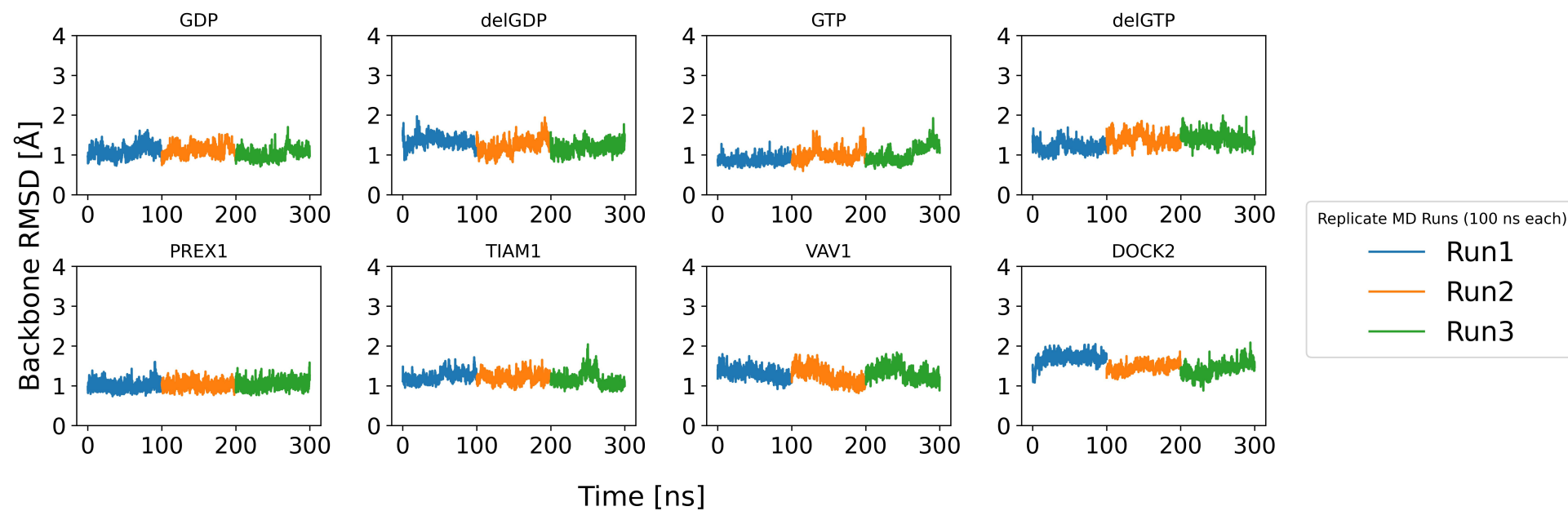


Figure S5: Time-evolution of RMSD of backbone atoms from Full Rac1 protein.



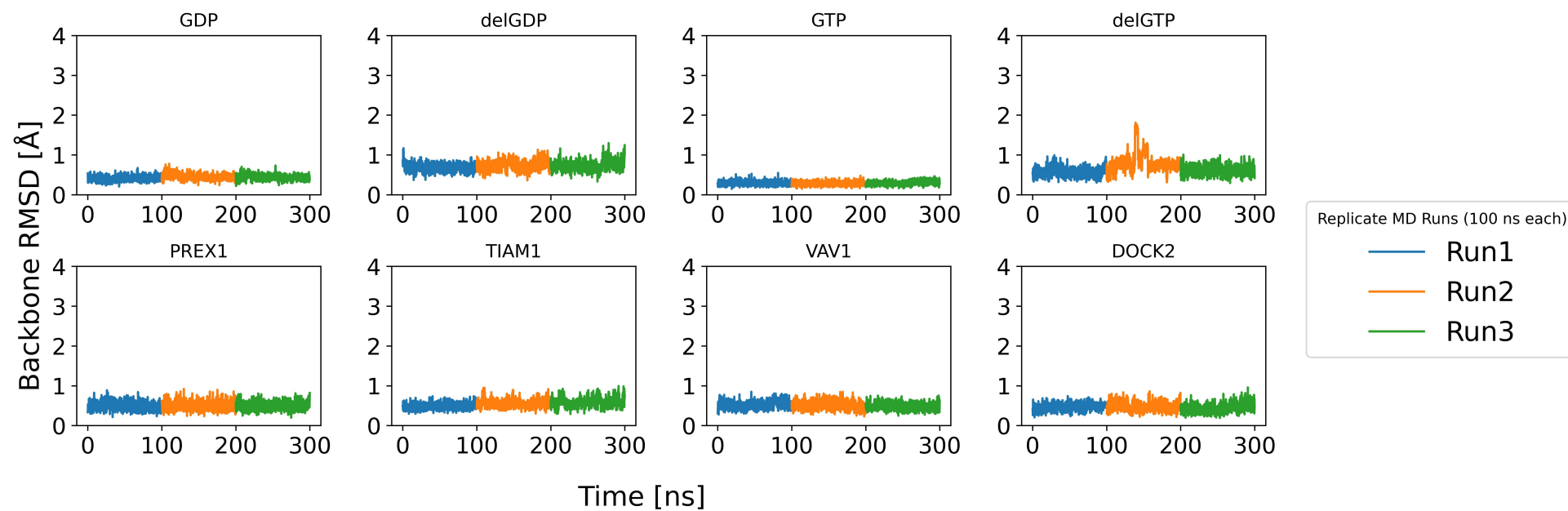


Figure S6: Time-evolution of RMSD of backbone atoms from the P-loop region.

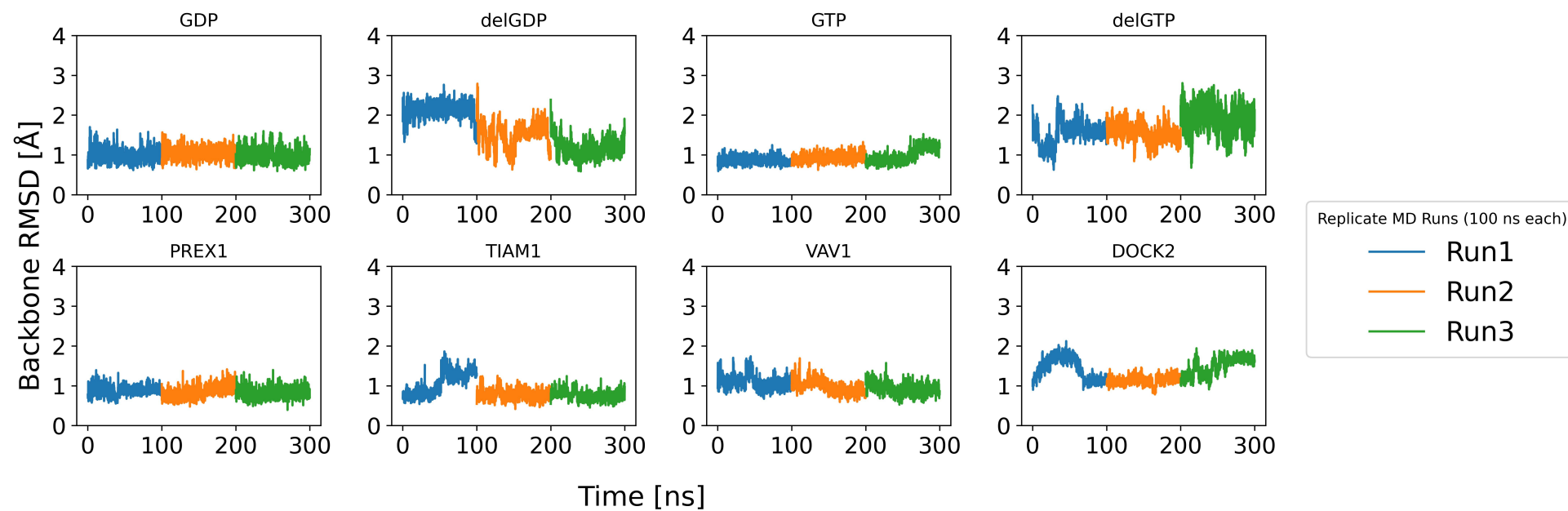


Figure S7: Time-evolution of RMSD of backbone atoms from the switch I region.

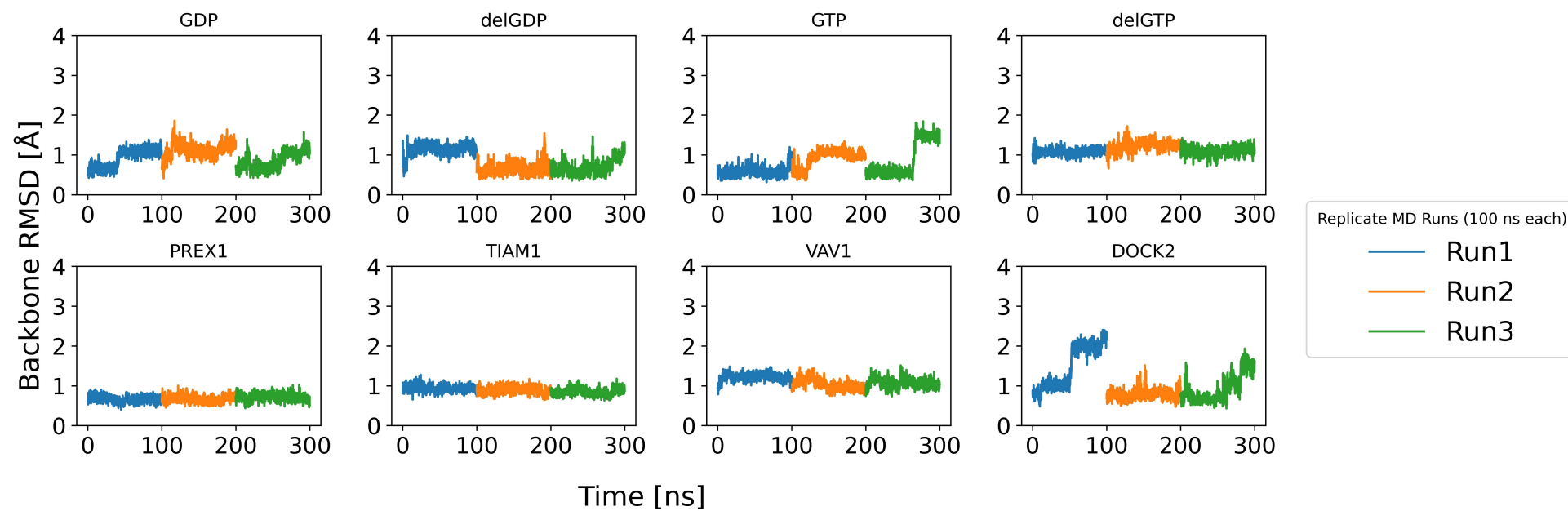


Figure S8: Time-evolution of RMSD of backbone atoms from the switch II region.